

Hearing the regularity of a diffusion coefficient: a graded local Weyl law with a constructive converse

Felix Robles Elvira (ORCID: 0009-0009-2017-4394; independent researcher)

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Abstract

The on-diagonal heat kernel of $L = -\partial_x(c(x)\partial_x)$ carries a local Weyl remainder $r_t(x) = K_t(x, x)\sqrt{4\pi c(x)t} - 1$. We prove *only* the smooth-class leading coefficient exactly — $r_t(x) = a_1(x)t + o(t)$ (remainder $O(t^2)$ for $c \in C^4$) with $a_1 = -c''/4 + (c')^2/(16c)$, via a Liouville reduction to a Schrödinger operator — and give a window heuristic with systematic fixed-seed evidence that the cusp-local decay rate of r_t *grades* the regularity of the coefficient at a symmetric cusp: a *symmetric* C^α cusp reads as $t^{\min(1, \alpha/2)}$ across the full range $\alpha \in (0, 2]$ and a coefficient in the Favard/saturation class $\{c : c'' \in L^\infty\}$ as t^1 (Conjecture G; the uniform two-sided *proof* for merely-Hölder c is the one open obligation, its route stated). We then study the converse: does a full-rate reading force membership in the regularity class? For the *practically available* (coefficient-free) form of the detector, as a statement over any *finite window of scales* the converse is **false**, by homogenization: a coefficient with fine (analytic, but large-norm) microstructure reads, at every scale above its cell size, as a smooth *effective* medium, the fitted coefficient agreeing with the classical harmonic mean to 0.6%. Every such impostor carries its own expiry scale, breaking away at $O(1)$ below the crossover $t \sim \delta^2$. The repaired statement is therefore scale-quantified — full-rate decay *at all scales* conjecturally characterizes the class (Conjecture C below; the frame is the semigroup characterization of Hölder–Besov spaces, the proof obligation — uniform two-sided parametrix bounds for merely-Hölder coefficients — is stated and not discharged) — and it is probed constructively: from detector data alone we recover the Hölder exponent to ± 0.03 (and, in a de-circularized variant that never consults the true coefficient, to comparable accuracy), locate the singularity to 7×10^{-4} (twenty times finer than the probing kernel’s width), verify a two-sided near-linearity of the signal in the cusp amplitude (mildly sublinear: signal/amplitude drifts from 0.073 to 0.065 over an $8\times$ range — the honest spread $\times 1.12$), and classify smooth / cusp / scale-break cases blind at 12/12, with the principled rate-floor threshold dictated by the graded law (admissible exponents lie in $(0, 1]$, so a non-decaying reading within the window belongs to no Hölder class at those scales and is an impostor signature) and the residual and smooth thresholds tuned on these same cases — so the 12/12 is not held-out generalization. An adversarial study shows a genuine cusp cannot be hidden, within a search-bounded smooth-plus-oscillatory dressing class, across the scale ladder. A load-bearing numerical lemma is included: the lattice artifact of the discrete kernel admits

an exact constant-coefficient rescaling baseline, $K_t(c) = K_{ct}(1)$, whose subtraction improves the detector floor from 10^{-2} to 10^{-7} — without it, every measurement above is noise. Scope: one dimension; the multidimensional and Lorentzian ports are discussed.

1. Introduction

Weyl asymptotics and heat-kernel expansions are the classical bridge between spectrum and geometry [1, 2, 3]. For smooth coefficients the on-diagonal kernel of $L = -\partial_x(c \partial_x)$ on the circle satisfies $K_t(x, x) = (4\pi c(x)t)^{-1/2}(1 + a_1(x)t + \dots)$; the *local Weyl remainder*

$$r_t(x) = K_t(x, x)\sqrt{4\pi c(x)t} - 1$$

then decays linearly in t , $r_t(x) = a_1(x)t + o(t)$ (the remainder sharpening to $O(t^2)$ for smoother c ; see Proposition 1.0), with an explicit first coefficient.

Proposition 1.0 (smooth-class leading term, exact coefficient). For $c \in C^2$ (remainder $o(t)$; sharpening to $O(t^2)$ when $c \in C^4$),

$$a_1(x) = -\frac{c''(x)}{4} + \frac{(c'(x))^2}{16c(x)}.$$

Proof. The Liouville substitution $y = \int^x c^{-1/2}$, $v = c^{1/4}u$ conjugates L to the Schrödinger operator $\tilde{L} = -\partial_y^2 + Q$ with potential $Q = c''/4 - (c')^2/(16c)$ (computed in the x -variable). Since $U : u \mapsto c^{1/4}u$ is unitary from $L^2(dx)$ to $L^2(dy)$ (as $dx = c^{1/2}dy$), conjugation gives $\tilde{K}_t(y, y') = c(x)^{1/4}c(x')^{1/4}K_t(x, x')$, hence on the diagonal $K_t(x, x) = c(x)^{-1/2}\tilde{K}_t(y, y)$. The flat Schrödinger expansion $\tilde{K}_t(y, y) = (4\pi t)^{-1/2}(1 - Qt + o(t))$ — the standard on-diagonal heat coefficient for $-\partial_y^2 + Q$ in one dimension (no curvature term in 1d; immediate on the diagonal from the Duhamel expansion of $e^{-t(-\partial_y^2 + Q)}$ about the free heat semigroup, whose first order contributes $-tQ$; the next, $O(t^2)$, term needs $Q \in C^2$, i.e. $c \in C^4$) — then gives $r_t(x) = -Q(x)t + o(t)$, sharpening to $O(t^2)$ for $c \in C^4$. $\square$

The continuum coefficient is thereby pinned *symbolically* — in particular the leading c'' coefficient is exactly $-1/4$ — and this is the value the symbolic reduction *certifies* at smooth points (the numerical detector reads the decay **rate**, not this pointwise value; see just below). The detector's demonstrated *numerical* capability is the **decay rate** of r_t (Section 6), not pointwise recovery of a_1 : the lattice baseline (Lemma 2.1) cancels the discretization artifact only to first order in the variation of c , leaving a residual comparable to the $O(t)$ continuum signal itself, so $a_1(x)$ is certified here through the symbolic reduction rather than read off pointwise. When c is merely Hölder the expansion fails and a_1 ceases to exist, and the natural question — in the spirit of "can one hear the shape of a drum" [4] but local and quantitative — is what the decay of r_t *hears*.

Conjecture G (the graded law, forward direction) — and its epistemic status. For c with a *symmetric* C^α cusp, r_t measured at the cusp on a locally-flat profile decays as $t^{\min(1, \alpha/2)}$ (the global $\sup_x |r_t|$ realizes this only for $\alpha \leq 1$; above, it is contaminated and saturates near 0.6, see Section 6). The mechanism is exact at leading order: the frozen-coefficient parametrix gives $r_t(x) = (\kappa/c(x)) \int G(u) [c(x + \sigma_t u) - c(x)] du + O(t)$ with $\sigma_t = \sqrt{2c(x)t}$ and $\kappa = -1/4$ (the constant of Proposition 1.0) — the remainder is a Gaussian-window modulus of continuity of

c at scale $\sqrt{t}$, so a C^α cusp reads as $t^{\alpha/2}$ and the saturation class as t^1 . (This single-window estimate captures the cusp/modulus scaling: for the *even* singular part of c the odd, c' -type contribution integrates to zero against the symmetric window, leaving the $-c''/4$ piece alone; the $(c')^2/(16c)$ piece of the full smooth a_1 is a higher-order parametrix correction it does not resolve. At a merely-Hölder cusp ($\alpha \leq 1$) c' need not exist, so this is purely an even-singular-part / window-modulus statement, not a claim about a vanishing derivative.) The $t^{\alpha/2}$ reading is the scaling of the *even (symmetric)* singular part of c at the cusp; the @cusp receipts and Conjecture G's leading-order claim are for cusps whose local self-similar (symmetric) structure dominates, where the single-window integral $\int G(u) |x|^\alpha du$ scales exactly as $\sigma_t^\alpha = t^{\alpha/2}$ for every $\alpha \in (0, 2)$. A one-sided C^1 cusp ($1 < \alpha < 2$) instead reads the larger of $t^{\alpha/2}$ and the local t^1 smooth background. The **smooth-class endpoint is a theorem** (Proposition 1.0); the graded law for non-smooth c is *numerically confirmed* (fixed-seed) across the *full* range $\alpha \in (0, 2]$ by the receipts of Section 6 (cusp-measured rates tracking $\min(1, \alpha/2)$ to within 0.01 at the cusp on flat-top profiles, and ~ 0.05 on non-flat-top profiles where plateau width limits the fit, at $\alpha = 0.5, \dots, 2.0$); the remaining obligation — a uniform two-sided parametrix bound proving the $t^{\alpha/2}$ rate is *exact* (both-sided) for every merely-Hölder c — is the named open core, with the route (Duhamel plus the heat-semigroup Hölder–Besov characterization [6]) laid out in Section 4 and not discharged. Here and throughout, "smooth" as a detector verdict means the **saturation class**: the $a_1(x)t$ term caps the readable rate at t^1 , and the first-order semigroup saturation heuristic suggests the $O(t)$ class should be the **Favard class** $\{c : c'' \in L^\infty\}$ ($C^{1,1}$ -type) — which we adopt as the meaning of the smooth verdict, the precise identification sharing Conjecture G's undischarged proof obligation — *strictly smaller* than the Zygmund-type $B_{\infty,\infty}^2$ (whose t^1 reading would need a second-order characterization, $(I - T_t)^2$) — and regularity beyond the saturation class is invisible to this detector.

The converse and its correction (main contribution). Does a rate-1 reading certify saturation-class membership? For the practically available detector — the coefficient-free ratio form of Section 5, since the oracle form below needs the answer as input — over a finite window of scales, no: Section 3 gives the homogenization impostor — oscillatory microstructure c_δ reads, for $t \gg \delta^2$, as the *smooth* effective medium with the harmonic-mean coefficient (the classical 1d homogenization limit [5]), so that detector does not merely miss the fine structure, it performs the homogenization. We stress the logical type of this counterexample: c_δ is analytic, with finite (merely enormous) class norms, so the *asymptotic* ($t \downarrow 0$) converse is never violated — what fails is every fixed-window version of it, which is the version any finite measurement uses. The corrected converse must therefore be **scale-quantified**: full-rate decay at *all* scales (Conjecture C: in the semigroup-characterization frame [6], with the same undischarged proof obligation as Conjecture G). It is probed **constructively**: Section 5 demonstrates the inversion — class, position, amplitude — with blind classification at 12/12 including the impostor class, whose signature (a non-decaying or scale-broken reading within the window) is exactly what the graded law's exponent range $(0, 1]$ forbids for genuine Hölder classes at those scales.

Why this matters beyond spectral theory. If Conjectures G and C hold, the diagnostic upgrades to a characterization: "the coefficient is in the saturation class" and "the detector reads rate 1 at all scales" become interchangeable at stated scope, with the homogenized media correctly *classified* (scale-break class) rather than misread. In applications where an effective geometry emerges from microstructure, by-scale classification is the physically meaningful statement.

2. Setting and the corrected detector

Unit circle, $L = -\partial_x(c\partial_x)$ with $0 < c \in L^\infty$, symmetric finite differences at $N = 512$, exact eigensolve; scale ladder $t \in [2.5 \times 10^{-4}, 6 \times 10^{-3}]$ (probing widths ≥ 8 grid cells).

Lemma 2.1 (lattice baseline). For constant c , the discrete kernel satisfies $K_t^{\text{disc}}(c) = K_{ct}^{\text{disc}}(1)$ exactly. Hence the lattice artifact admits the baseline $g(s) := r_s^{\text{disc}}(\cdot; c \equiv 1)$ and the **corrected detector** $\tilde{r}_t(x) = r_t(x) - g(c(x)t)$ removes it to first order in the variation of c .

Receipt: floor improvement from 1.5×10^{-2} to 6×10^{-7} ; without the baseline subtraction the "measurements" are pure discretization — smooth coefficients read rate -0.02 where the law says 1.0 . We flag this prominently because any implementation of local-Weyl numerics faces it. The correction is first-order in the variation of c : the uncorrected artifact is $O(t)$, and the subtraction leaves a second-order residual of relative size $O((c'/c)^2(dx/\sqrt{t})^p)$. The quoted 6×10^{-7} floor is the floor for the *rate* measurement (the use to which the detector is put in Sections 5–6); it certifies the slope, not pointwise recovery of $a_1(x)$, for which the same residual is comparable to the signal.

3. The finite-window converse is false: the homogenization impostor

Observation 3.1 (impostor; classical limit plus receipts — stated for the right detector). The $t \gg \delta^2$ harmonic-mean limit below is classical homogenization [5], cited not re-proved; the new content is the detector-level quantification (crossover, $O(1)$ breakaway, 0.6% fidelity). Let $c_\delta(x) = 1 + a \sin(2\pi x/\delta)$. For $t \gg \delta^2$ the diagonal kernel is spatially flat and the **coefficient-free reading** — the fitted coefficient $\hat{c} = 1/(4\pi t K_t(x, x)^2)$, equivalently the ratio detector of Section 5 — reads c_δ as a smooth medium whose reported coefficient is the harmonic mean $\bar{c}_H = (\int c^{-1})^{-1}$; for $t \ll \delta^2$ the reading breaks away at $O(1)$.

Which detector is fooled. The impostor fools the *coefficient-free* detector, not the oracle form. The oracle detector $\tilde{r}_t$ of Lemma 2.1 normalizes by the *true* $\sqrt{c(x)}$, which oscillates at $O(1)$ against the homogenized (flat) kernel, so it reads $\sup |\tilde{r}_t| \approx \sqrt{(1+a)/\bar{c}_H} - 1 \approx 0.41$, *non-decaying at all scales above the crossover* — it is never fooled, and that non-decaying signature is precisely how the blind classification of Section 5 catches the microstructure (fitted rate ≈ -0.07 there). The impostor phenomenon — and hence the finite-window failure of the converse — therefore belongs to the *coefficient-free* detectors, which are also the only ones a real measurement has. The narrative of Sections 1 and 5 is stated for those.

Status of the statement. The $t \gg \delta^2$ regime is a consequence of classical periodic homogenization [5] (G-convergence of L_δ to the harmonic-mean operator, hence locally uniform convergence of the kernels); we do not re-prove it. What we add, and verify by machine rather than prove, is the *detector-level quantification*: the crossover location, the $O(1)$ breakaway size, and the fidelity of the reported coefficient at finite δ .

Receipts ($\delta = 1/32$, $a = 0.6$): fitted effective coefficient 0.8045 vs $\bar{c}_H = 0.8000$ (within 0.6%; because the profile here is deterministic, the comparison carries no realization-versus-ensemble gap — unlike the random-coefficient example of Section 6, where that gap must be accounted for — and the arithmetic mean 1.0 is excluded at 25%); oracle detector above crossover: $\sup |\tilde{r}| \approx$

0.41, non-decaying, as the correction paragraph states; below crossover both readings break at $O(1)$ ($\sup |\tilde{r}| = 0.52$).

Note again the type of the example: c_δ is analytic. The impostor does not contradict the asymptotic converse; it shows that the converse *as restricted to any finite scale window* — the only version a finite-resolution measurement can instantiate — is false, and that the failure is physical (effective-medium formation), not pathological. The failure mode dictates the fix:

Conjecture C (corrected converse). If the coefficient-free detector reads rate $1 - O(t)$ decay — *at every scale* $t \downarrow 0$, then c belongs to the Favard/saturation class $\{c : c'' \in L^\infty\}$ of Section 1; more generally the all-scale rate $t^{\alpha/2}$ characterizes $B_{\infty,\infty}^\alpha$ membership for $\alpha \in (0, 1)$ (the Hölder range, where first-order semigroup characterizations [6] apply). *Both directions of this equivalence remain conjectural:* the forward bound is Conjecture G, the converse is the two-sided window estimate of Section 4, and the two share a single undischarged proof obligation — uniform two-sided parametrix bounds for merely-Hölder c — which is open and stated, not papered over.

4. The two-sided mechanism

The kernel window bounds the local oscillation of c at scale $\sqrt{t}$ from both sides. The load-bearing receipt is the **amplitude sandwich**: for a $C^{0.5}$ cusp of amplitude η at fixed scale,

eta:	0.2	0.4	0.8	1.6
signal:	0.0146	0.0287	0.0556	0.1046
signal/eta in [0.0654, 0.0730] (two-sided, spread x1.12)				

A detector zero at scale t would force the *symmetric, sign-definite* part of the oscillation to vanish at scale $\sqrt{t}$; promoting this to a genuine modulus-of-continuity bound at every scale — Besov membership — requires controlling cancelling oscillations, which is precisely the two-sided parametrix estimate left open in Conjecture C. Conversely the forward graded law bounds the signal above by the class norm. At the stated 1d scope this is the mechanism behind Conjecture C for $\alpha \in (0, 1)$ (the Hölder range; at $\alpha = 1$ the all-scale rate $t^{1/2}$ tracks the Zygmund class $B_{\infty,\infty}^1$, not Lipschitz, and we make no equivalence claim there) plus the saturation endpoint.

5. The constructive inversion

From detector data alone:

- **Class:** $\hat{\alpha} = 2 \times$ (fitted rate), capped at smooth. Across random positions and amplitudes ($\alpha \in \{0.5, 1.0\}$): $\max |\hat{\alpha} - \alpha| = 0.03$.
- **Position:** $\arg \max_x |\tilde{r}_t(x)|$ at the smallest valid scale lands within 7×10^{-4} of the cusp (kernel width 1.6×10^{-2}): the inversion is local.
- **Blind classification** over 12 cases (5 smooth random Fourier media, 5 cusps with random parameters, 2 microstructures): decision rule — fit residual > 0.25 *or* rate < 0.1 declares SCALE-BREAK; rate ≥ 0.85 declares SMOOTH; otherwise CUSP with $\hat{\alpha}$. Result: 12/12. Threshold honesty: the rate *floor* (0.1) is principled, not tuned — graded exponents live in $(0, 1]$, so a non-decaying reading (microstructure enters at fitted rate ≈ -0.07 with small residual — a power law, but an inadmissible one) belongs to no Hölder class — while the

residual threshold (0.25) and the smooth threshold (0.85) *are* tuned choices, and the 12/12 should be read with that asymmetry in mind.

The three-way output — smooth / Hölder / scale-break — is the correct taxonomy: the third class is *physics* (effective media), not noise, and the constructive converse classifies it instead of being fooled by it.

De-circularizing the detector. A fair objection: $\tilde{r}_t$ contains $\sqrt{c(x)}$ and $g(c(x)t)$ — the inversion above consults the very coefficient it claims to probe. Two remarks. First, the naive repair (estimate $\hat{c}$ from K_t at the smallest window scale and substitute) *fails by construction*: the detector then vanishes at the calibration scale and the decay fit measures the calibration, not the roughness — we ran it, and the fitted exponents pin at the cap; we record this as a warning to implementers. The correct repair is the **ratio detector**

$$s_\lambda(x, t) = \frac{K_{\lambda t}(x, x) \sqrt{\lambda} (1 + g(\hat{c}(x)t))}{K_t(x, x) (1 + g(\hat{c}(x)\lambda t))} - 1,$$

in which the unknown $\sqrt{4\pi c(x)t}$ normalization cancels *exactly* between numerator and denominator; a crude $\hat{c}$ (smallest-scale estimate) survives only inside the lattice correction g , where its $O(t^{\alpha/2})$ error multiplies a term that is itself $\lesssim 10^{-3}$. The roughness signal survives with the same exponent: $\sup_x |s_\lambda| \sim t^{\alpha/2}(\lambda^{\alpha/2} - 1)$. Receipt ($\lambda = 4$, three random cusp positions/amplitudes per class, both grids):

alpha = 0.5:	fitted rates 0.224 - 0.237	(law: 0.25)
alpha = 1.0:	fitted rates 0.489 - 0.495	(law: 0.50)
smooth:	rate 0.944 / 0.945	(law: ~1)

The inversion needs no oracle access to c .

6. The adversarial wall and honest limitations

Adversarial study. Fix a genuine $C^{0.5}$ cusp of amplitude η and minimize the detector excess over smooth-plus-oscillatory dressing (nine parameters, local search):

smooth control excess:	0.072
eta = 0.2 / 0.4 / 0.8:	0.306 / 0.538 / 0.929 (minimized)

Monotone in η , never approaching the control: in this dressing class a cusp cannot be hidden from the ladder. *Limitations:* the dressing class is small and the search local — the wall is search-bounded; a certificate version (sum-of-squares over the dressing class) is the natural upgrade.

The $\alpha \in (1, 2)$ band, resolved. The graded rate holds across the *whole* range once the measurement is taken at the right place. The global statistic $\sup_x |r_t|$ is contaminated for $\alpha > 1$ by the smooth background away from the cusp and by any boundary corner of the test coefficient: a gentle (C^{1+}) cusp contributes only a weak $t^{\alpha/2}$ signal, which the background's t^1 contribution — large in absolute constant — masks, pinning the *apparent* global rate near 0.5–0.6 for every $\alpha \geq 1$. Measuring r_t **at the cusp** on a flat-top coefficient (the test profile $\equiv 1$ on a plateau wider than $\sqrt{t}$ around the singularity, smoothly cut off before any other feature) removes the contamination and recovers the law cleanly through the band:

alpha	0.50	0.75	1.00	1.25	1.50	1.75	2.00	
predicted	0.25	0.375	0.50	0.625	0.75	0.875	1.00	(min(1,a/2))
measured@cusps	0.243	0.365	0.491	0.619	0.745	0.872	0.999	
measured(sup)	0.243	0.365	0.557	0.611	0.606	0.601	0.595	(contaminated)

The `measured(sup)` row, saturating near 0.6, is the artifact; `measured@cusps` tracks $\min(1, \alpha/2)$ to within 0.01 across the full $(0, 2]$. The forward graded law is therefore *numerically confirmed* (fixed-seed) for all $\alpha \in (0, 2]$, not only $\alpha \leq 1$; the open obligation is the two-sided *proof*, not the band.

Grid refinement. The headline rates are not lattice artifacts: refining $N = 512 \rightarrow 768$ (a $1.5\times$ finer grid) leaves every fitted rate unchanged to three decimals — 0.232/0.494/0.969 vs 0.232/0.494/0.970 for $\alpha = 0.5/1.0/\text{smooth}$ — and the de-circularized rates of Section 5 move by < 0.013 .

Dimension: the 2d port, executed. The parametrix mechanism and the Besov frame are dimension-generic, and we verify the port on the 2-torus ($L = -\nabla \cdot (c\nabla)$, 40^2 grid, the same exact rescaling baseline — valid in any dimension):

```
smooth rate 0.880 (law: 1.0, window-limited);  cusps rates
0.197 / 0.452 at alpha = 0.5 / 1.0 (law: 0.25 / 0.50);
position recovery to 0.015 (kernel width 0.035).
```

```
THE 2d IMPOSTOR has its own exact classical anchor: for a
statistically isotropic symmetric two-phase medium the
effective coefficient is the GEOMETRIC MEAN (Dykhne/Keller
duality [11]). Random checkerboard (c1, c2) = (0.5, 2.0):
kernel-reported effective c = 1.0057 vs geometric mean 1.0000
(0.6%). Quantity bookkeeping, stated: 1.0000 is the ENSEMBLE
Dykhne/Keller target (symmetric fractions); the random
realization has high-phase fraction p-hat ~ 0.47 (low-phase
fraction ~ 0.53), whose own
arithmetic mean is the quoted 1.205 (not the ensemble 1.25),
and realization fluctuation contributes at the same order as
the quoted 0.6% - so the receipt is "ensemble target matched
to within realization noise", with the arithmetic mean
excluded at 20%. Below the cell scale the reading breaks
at 0(1).
```

The detector thus performs the *correct, dimension-specific* homogenization in 2d (geometric-mean target matched to 0.6%), with the cusps rates tracking $\min(1, \alpha/2)$ up to the coarse-grid undershoot (0.197/0.452 vs 0.25/0.50 at 40^2) — evidence that the corrected (all-scale) converse plausibly extends beyond 1d, with a quantitative 2d study left as a sequel. The general- d theorem with anisotropic coefficient classes, and a Lorentzian port (Euclidean-first through Osterwalder–Schrader rotation), remain the named sequels.

7. Related work

Heat invariants and local Weyl expansions for smooth coefficients are classical [1, 2, 3]; regularity detection via semigroup approximation rates is implicit in the Besov characterization literature [6]; homogenization of oscillatory coefficients is textbook [5]. Two adjacent literatures must be

distinguished from what is done here. *Inverse spectral theory* (Borg [7]; Gel’fand–Levitan [8]; Marchenko [9]) recovers the coefficient itself from full spectral data (spectrum plus norming constants); we use far less input — the decay *rate* of the on-diagonal kernel — and obtain a correspondingly coarser output, a regularity class with localization, which is why the method survives non-smooth coefficients where the classical machinery assumes regularity. *Aronson-type Gaussian bounds* [10] guarantee that K_t and hence the detector are well-defined for merely measurable uniformly elliptic c , which is what licenses evaluating $\tilde{r}_t$ on every class discussed, including the microstructures. The contributions here are (i) the graded *local* law (read at the cusp; the global sup saturates above $\alpha = 1$) with receipts at non-smooth classes, (ii) the impostor proposition identifying the precise failure (and physical meaning) of the finite-window converse, (iii) the constructive inversion with the principled three-way decision rule and its de-circularized variant, and (iv) the adversarial formulation of detector rigidity.

Reproducibility

Fixed-seed scripts regenerate every number: the baseline lemma and floor receipts; the graded rates; the homogenization receipts; the inversion and blind classification; the adversarial walls. Code and canonical outputs accompany the submission.

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